

New material for seat recliner

Specification and test recommendation



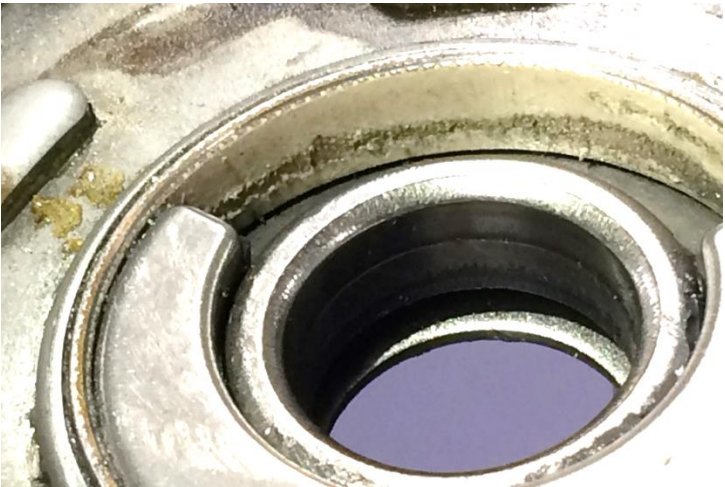
UNLOCKING
YOUR
ENGINEERING
PUZZLES

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Material Specification

■ Basic material performance is ideally as same as PRO100E

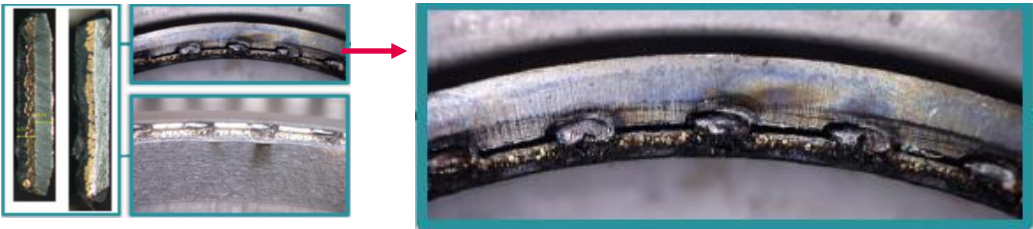
Maximum admissible specific bearing load at RT	static dynamic SG PL 0044	N/mm ² N/mm ²	400 200
Coefficient of friction at RT measured on steel with ≥ 58 HRC	at 4.8 N/mm ² and 0.058 m/s at 70 N/mm ² and 0.0065 m/s SG PL 0003		0.21 0.13
k factor at RT measured on steel with ≥ 58 HRC	at 4.8 N/mm ² and 0.058 m/s SG PL 0003	10 ⁻⁶ mm ³ /Nm	0.18
Deformation under load	23°C, at 100 N/mm ² , 1h SG PL 0015	μm	≤ 5
Rate of heat transfer	SG PL 0001	W/m ² K	5000
Service temperature range (to be validated in the application)	permanent intermittently	°C	-200 - +260 290
Coefficient of linear thermal expansion	SG PL 0020	10 ⁻⁵ K ⁻¹	4
Surface related volume resistance	SG PL 0038	Ωcm ²	≥ 10 ¹²



Reference: **New** recliner unit from after market. PRO100E is used(MAZDA). The sliding layer 'broke" not worn. Thinner and harder sliding layer is better to avoid this failure mode??

➡ Application doesn't need +260°C but better to meet this level for our new biz. Outside of recliner.

- Similar CoF gap between static/dynamic friction and wear resistance to PRO100E.
- Heat resistance
 - No damage or delamination due to welding heat (Up to 600°C).
 - FORVIA welding back metal to fix it to the housing (spot welding).



- Need chamfering capability of back metal side(assembly of bearing) and sliding layer side(assembly of wedges).
- Flanging capability without damage or crack.
- Back metal
 - Similar deformation under load performance to PRO100E.
- Thickness
 - 0.50mm, 1.00mm, 1.50mm

Material Specification

■ Cost

- Minimum half of PRO100E material cost with no additional forming cost (same level of production cost).
 - Target modern recliner systems that dominated by DU type products(0.10 - 0.20 euro/pc).

■ Forming

- Tangling free
 - Closed gap without puzzle gap and welding that cause of bump after assembly.
 - New forming technology/concept is needed.



Free state condition of competitor bush

■ Outer diameter is controlled by delta Z measurement of ISO3547-2.

- Press fit application. Specification of retention force is...
 - >1000N at $\phi 35 +0.021/0$ mm housing (TF-METAL)
 - >900N at $\phi 40.86 \sim 40.90$ and $Ra 0.1 \sim 0.3$ housing (IMASEN)
 - * Normally controlled by delta Z measurement.

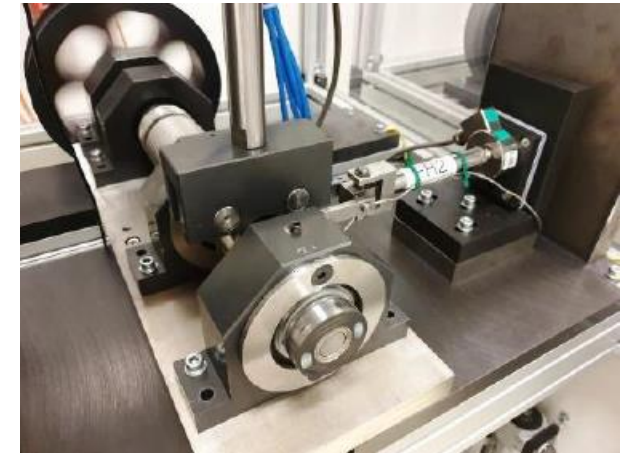
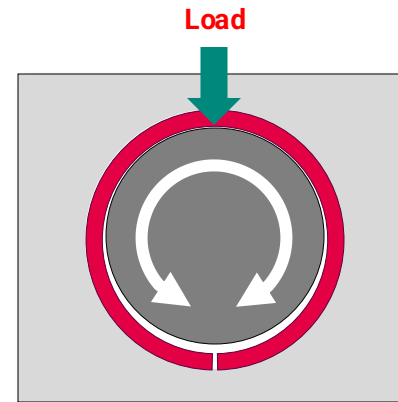
in Anlehnung an DIN ISO 3547 Teil 2 in style of DIN ISO 3547 part 2								
Legende: 1 Stoßfuge joint 2 Prüfaufnahme master-gauge	Prüfaufnahme und Einstellhorn master-gauge and adjusting-mandrel	$d_{ch,1} = d_{ch,2} =$ 33,067						
	Prüfkraft für Buchse test-force for bush	$F_{ch} =$ 1500 N						
	Grenzwert für Delta-z limit of Delta-z	Delta z = 0 und -0,065						
Innen-Ø Prüfung / test the inside-Ø Prüfung in Lehre Ø 33,07 test in gauge Ø 33,07								
<table border="1"> <tr> <td>GO-Pin</td><td>Ø 31,06</td><td>js5 < 100N</td></tr> <tr> <td>NOGO-Pin</td><td>Ø 31,15</td><td>js5 < 100N</td></tr> </table>			GO-Pin	Ø 31,06	js5 < 100N	NOGO-Pin	Ø 31,15	js5 < 100N
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Example

Recommendation of test method

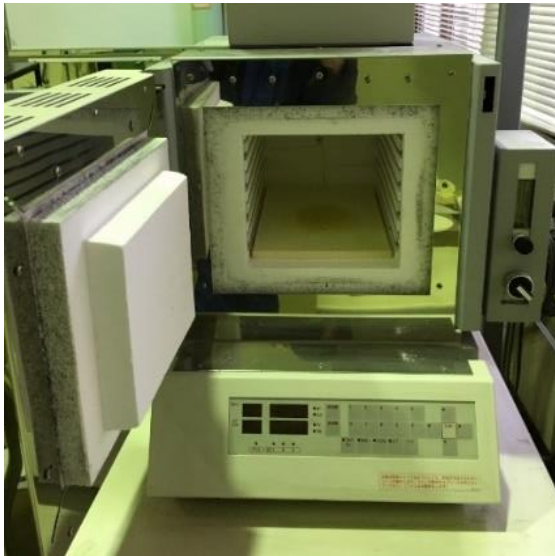
- Linear bearing tester (HEIDON or Yoke tester?)
 - Surface pressure: 110MPa
 - Speed: 35mm/sec
 - Cycle: 33,750 cycles
 - $\pm 52^\circ$ movement = CW/CCW 5 shaft rotations each
 - 6750 durability cycles x 5 rotations = 33,750 cycles
 - Material comparison
 - New material vs PRO100E, OILES Drymet LF and GGB DU

- JBT Generation 2 (Willich)
 - Estimate based on above.

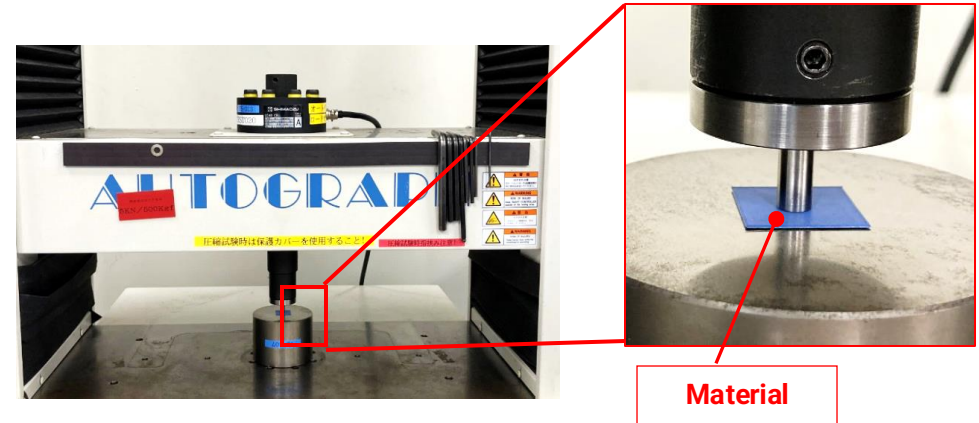
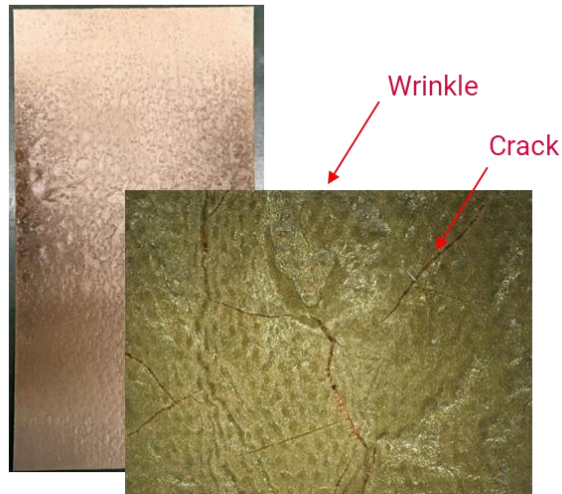


Recommendation of test method

- Compression test
 - Measure deformation under up to 400MPa
- Heat resistance
 - Heat chamber: 600°C x 1 minutes



600°C / 1min



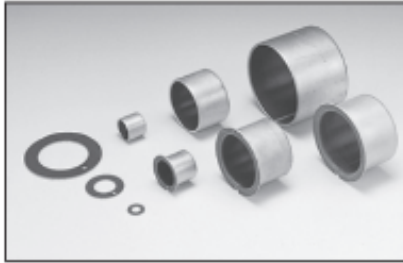
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Part number	Thickness		Plate size	
	Thickness	Tolerance	Width	Length
LFP-1080	1.0	-0.13 ~ -0.03	80	500
LFP-1590	1.5	-0.13 ~ -0.03	90	500
LFP-20100	2.0	-0.13 ~ -0.03	100	500
LFP-25100	2.5	-0.13 ~ -0.03	100	500

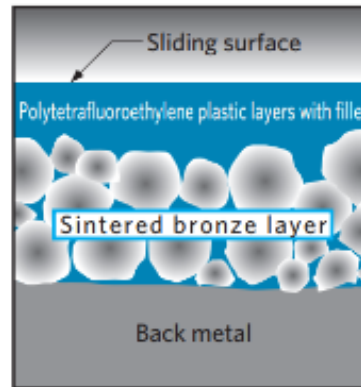
Reference - Competition

Oiles Drymet LF Polytetrafluoroethylene plastic multi-layer bearings with back metals



Feature

- Serviceable without the need for lubrication. Thin and lightweight for compact design.
- Demonstrate low coefficient of friction and superior load performances in high-load, low-speed conditions.
- Usable at the high PV values even in the middle- and high-speed zones when lubricated.
- Usable in wide temperature ranges from low to high temperatures. Has superior chemical resistance.
- Features superior dimensional stability, mechanical strength, and thermal conductivity.
- The standard products and plate materials are available in various sizes.



image

Service range

Lubrication condition	Dry
Service temperature range °C	-200~+280
Allowable max. pressure P N/mm ² {kgf/cm ² }	49.0 (137) {500 (1,400)}
Allowable max. velocity V m/s {m/min}	0.65 {39}
Allowable max. PV value N/mm ² · m/s {kgf/cm ² · m/min}	3.60 {2,200}

The values in parentheses are static bearing pressures, which are the bearing pressures in applications with no motion or very small motion ($\leq 0.0017\text{m/s}$ [0.1m/min]).

※ Allowable max. pressure of LFCF is 24.5N/mm^2 [250Kgf/cm²].

Mechanical properties

Tensile strength	JIS Z 2241	N/mm ² {kgf/cm ² }	380 {3,875}
Elongation	JIS Z 2241	%	27
Hardness	JIS Z 2244	HV	107

※ The values shown above are typical values, not the standard values.

※ The values shown above are values of back metal.

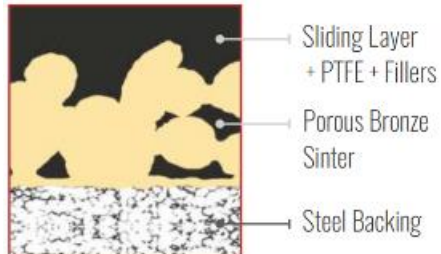
Reference - Competition

- GGB DU® : We can use distributor of GGB.



DU®

METAL-POLYMER ANTI-FRICTION
PLAIN BEARINGS



BEARING PROPERTIES		UNITS	VALUE
GENERAL			
Maximum load, p	Static	N/mm ²	250
	Dynamic	N/mm ²	140
Operating temperature	Min	°C	- 200
	Max	°C	280
Coefficient of linear thermal expansion	Parallel to the surface	10 ⁻⁶ /K	11
	Normal to the surface	10 ⁻⁶ /K	30
DRY			
Maximum sliding speed, U		m/s	2.5
Maximum pU factor		N/mm ² x m/s	1.8
Coefficient of friction, f			0.02 - 0.25*
OIL LUBRICATED			
Maximum sliding speed, U		m/s	5.0
Maximum pU factor		N/mm ² x m/s	5.0
Coefficient of friction, f			0.02 - 0.12*
RECOMMENDATIONS			
Shaft surface roughness, Ra	Dry	µm	0.3 - 0.5
	Lubricated	µm	≤ 0.05 - 0.40*
Shaft surface hardness	Unhardened acceptable, improved bearing life	HB	> 200

* Depending on operating conditions

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THANK YOU

ANY QUESTIONS?

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